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ABSTRACT

Traditional Fourier heat conduction fails to accurately describe thermal responses under rapidly varying or high-frequency heating conditions. In such scenarios, heat transfer exhibits non-Fourier behavior, where the heat flux does not instantly follow the temperature gradient. This deviation requires a more comprehensive model that incorporates relaxation times, representing the delay in the heat flux's response to temperature changes. The relaxation time of a system is often assumed to be linked to its phonon relaxation times or other intrinsic characteristic times, but a definitive method for its characterization, and a robust theoretical framework to describe them remains an open problem. By modeling a frequency-domain thermo-reflectance (FDTR) experiment taking into account non-Fourier heat transfer, we investigate the temperature distribution in materials with diverse relaxation times, capturing transient thermal profiles. The results demonstrate high sensitivity to relaxation-time parameters in FDTR experiments and provide a practical approach for characterizing the relaxation times in non-Fourier materials.

I. INTRODUCTION

Non-Fourier heat conduction is a branch of thermal physics that studies heat transfer phenomena that deviate from classical Fourier's law. In particular, non-Fourier heat conduction departs from the conventional Fourier's theory assumption[1] of linear proportionality[2] between the heat flux $\vec{q}$ and the temperature gradient ∇T. In scenarios involving materials with non-negligible thermal relaxation times or rapid temperature changes, the classical Fourier model falls short of capturing the transient behavior of heat propagation. To address these shortcomings, researchers have turned to alternative models that can consider phenomena, such as non-local effects,[3] thermal relaxation times,[4] and ballistic transport of energy carriers.[5]

One prominent framework is the Cattaneo–Vernotte (CV) equation.[6,7] The CV equation, introduced by Carlo Cattaneo[6] and P. Vernotte,[7] modifies the classical heat conduction equation. Unlike Fourier's instantaneous response assumption, the Cattaneo equation incorporates a finite thermal relaxation time τ. This relaxation-time parameter describes a characteristic time required for the heat flux to respond to changes in the temperature gradient or equivalently the characteristic time for a material to achieve thermal equilibrium after experiencing a perturbation. An essential consequence of introducing a relaxation time is that the corresponding differential equation describing the temperature profile becomes hyperbolic instead of the parabolic Fourier equation.[8,9] Thus, in principle, the behavior of the temperature field can be modulated via typical wave properties, such as scattering,[10] interference,[11] and dispersion relations specific to the system.[12–14] While the CV equation captures transient heat conduction, there are several other models that incorporate a relaxation time. For example, a closely related model is the dual-phase lag model, which introduces two relaxation times.[15,16] One relaxation time considers the heat flux response to changes in the temperature gradient, and another describes the response time of the temperature gradient to changes in the heat flux.[17,18] An extensive contemporary review of non-Fourier heat equations can be found in Kovács.[19]

Changes in the temperature profile due to the introduction of a response time[20–22] can have important implications in many fields, such as heating of biological tissues for hyperthermia therapy[23–27] and in micro- and nano-electronics.[5,17,25,28–30] Temperonic crystals made of a unit cell of two different slabs with wave-like temperature propagation also show the usual band-like structure common in other wave phenomena.[31–33]

The relaxation time is expected to vary depending on the material properties, geometry, and boundary conditions of the system under consideration. However, no theoretical model or experimental procedure exists to establish the value of τ for a given material. Nonetheless, several experiments have been conducted to explore and validate non-Fourier heat conduction models. This is common in biological tissues, such as muscle,[34–37] as well as in media with high inhomogeneities, such as sand and foams.[38]

Time-Domain Thermoreflectance (TDTR) and Frequency-Domain Thermoreflectance (FDTR) are widely used to measure the thermal properties of thin films. Both TDTR and FDTR are non-contact pump-probe techniques used to measure the thermal response of materials. Absorption of the pump laser induces a thermal source at the sample surface. Changes in the reflected light from the probe beam are used to calculate the temperature on the surface of the sample based on the temperature dependence of the surface reflectivity. Typically, in TDTR experiments, ultrafast lasers are used as the pump beam, and a probe pulse is time delayed with respect to the pump pulse using a mechanical delay stage. Taking advantage of ultrashort laser pulses and pump-probe detection techniques allows for thermal quantities to be measured with picosecond time resolution. On the other hand, in FDTR experiments, the thermal properties of materials are measured by applying a modulated heat source, typically a laser, and analyzing the material's thermal response with a probe laser as a function of modulation frequency. Two implementations for the FDTR technique can be used: one based on an ultrafast pulsed laser system and one based on continuous-wave lasers. The frequency modulation ranges depend on the signal to noise ratio and are usually in the order of 10 kHz and up to 200 MHz.[39,40] This technique has been instrumental in the study of nanoscale heat conduction,[41] complex materials, such as polymers,[42] superlattices,[43] and semiconductors.[44]

The typical use of FDTR data involves fitting for thermal properties so that the consideration of a model with relaxation time is needed in order to establish the relaxation time of materials. In this work, we theoretically investigate the viability of using frequency-domain thermoreflectance to determine non-Fourier heat behavior. FDTR has the advantage of providing measurements in frequency space, simplifying the comparison with the theoretical model. Furthermore, the high sensitivity in phase measurement and a resolution of a few millidegress[40] make FDTR measurements a good platform for exploring non-Fourier behavior. We predict the FDTR results of a particularly interesting case of an experimentally feasible sample consisting of layered materials making up a 1D thermal crystal.

II. FOURIER AND CATTANEO–VERNOTTE HEAT CONDUCTION EQUATIONS

For the sake of completeness, we briefly show the two heat conduction models. The Fourier and Cattaneo–Vernotte heat equations for the temperature $T(\mathbf{r}, t)$ are

$$\nabla^2 T(\mathbf{r}, t) - \frac{1}{\alpha}\frac{\partial}{\partial t} T(\mathbf{r}, t) = -\frac{1}{k} s_f(\mathbf{r}, t) \tag{1}$$

and

$$\nabla^2 T(\mathbf{r}, t) - \frac{1}{\alpha}\frac{\partial}{\partial t} T(\mathbf{r}, t) - \frac{1}{v^2}\frac{\partial^2}{\partial t^2} T(\mathbf{r}, t) = -\frac{1}{k} s_{cv}(\mathbf{r}, t), \tag{2}$$

where ρ is the mass density, c_v is the specific heat at constant volume, $\alpha = k/(\rho c_v)$ is the thermal diffusivity, k is the thermal conductivity, and $v^2 = \alpha/\tau$ is the CV velocity, with τ being the CV relaxation time. The sources $s(\mathbf{r}, t)$ for the Fourier and CV cases are related by

$$s_{cv}(\mathbf{r}, t) = s_f(\mathbf{r}, t) + \tau \frac{\partial s_f(\mathbf{r}, t)}{\partial t}. \tag{3}$$

The source feeds the energy conservation equation

$$\frac{\partial u(\mathbf{r}, t)}{\partial t} + \nabla \cdot \mathbf{q}(\mathbf{r}, t) = s_{f/cv}(\mathbf{r}, t), \tag{4}$$

with the thermal energy $u = c_v \rho T$; for Fourier's case, we have the heat flux equation $\mathbf{q}_f(\mathbf{r}, t) = -k\nabla T_f(\mathbf{r}, t)$, while for Cattaneo's, $\mathbf{q} + \tau\partial\mathbf{q}/\partial t = -k\nabla T(\mathbf{r}, t)$. In the limiting case $\tau \to 0$, Fourier's case is recovered. By taking the Fourier transform in the time-coordinate, the equations for the temperature profile are

$$\nabla^2 T(\mathbf{r}, \omega) - \sigma^2_{f/cv} T(\mathbf{r}, \omega) = -\frac{1}{k} s_{f/cv}(\mathbf{r}, \omega), \tag{5}$$

where $\sigma_f = (i\omega/\alpha)^{1/2}$, $\sigma_{cv} = \sqrt{i\omega/\alpha - (\omega/v)^2}$, and $s_{cv}(\mathbf{r}, \omega) = s_f(\mathbf{r}, \omega)[1 + i\omega\tau]$.

It is important to notice that the thermal conductivity κ and the CV relaxation time τ depend on the temperature. The temperature dependence depends on the material and the temperature range. At high temperatures, the relaxation time becomes negligible, and Fourier behavior is expected. At very low temperatures, the mean free path becomes longer, and rather than a diffusive phonon regime, we are in a ballistic regime.[45] Usually, the thermal conductivity and the relaxation time follow an inverse power law with temperature,[46] and this is $\kappa \sim T^{-m}$ and $\tau \sim T^{-n}$. For Si, the values[47] $m = 1.4$ and $n = 1.3$ and the high temperature[48] limit where it behaves as a Fourier material[49] is 500 K and for SiC at around 600 K with similar values for the exponentials n and m. In a typical FDTR setup, the system is at room temperature and the changes in the temperature are small enough so as to assume that the thermal conductivity and the relaxation time remain constant.

III. FREQUENCY-DOMAIN THERMOREFLECTANCE

Pump-probe FDTR involves a modulated laser source (pump) to heat the sample and a second laser (probe) to measure changes in the sample's reflectance in response to the thermal excitation, as depicted in Fig. 1. In this work, we consider two cases: First, a

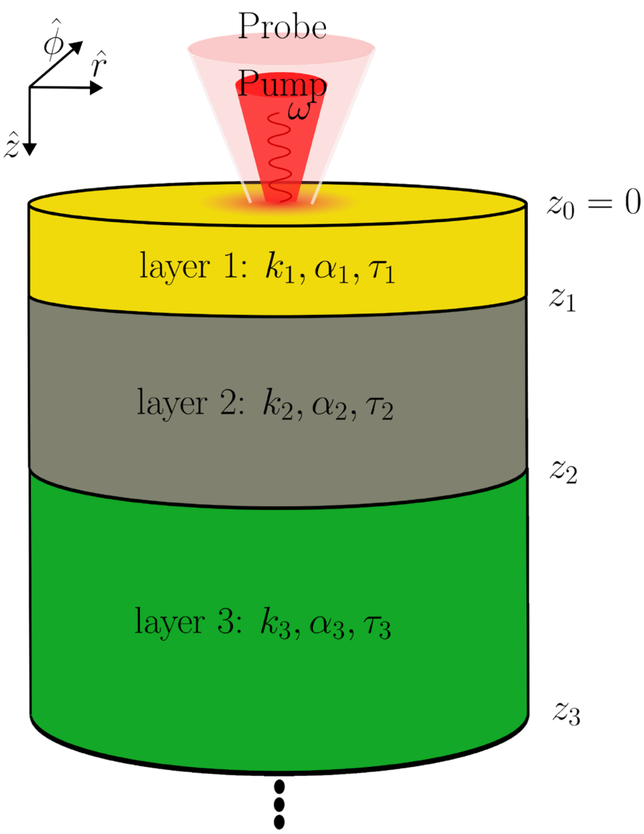

FIG. 1. Schematic of an FDTR setup. The temperature is measured indirectly through the reflectance $R(\lambda_{probe}, T(\mathbf{r}, \omega))$ of the sample, which is dependent on the probe wavelength λ_{probe} and the temperature at the surface $T(\mathbf{r}, \omega)|_{z=0}$.

half-space, which occupies the region $z > 0$, and then a laminated system like the one shown in Fig. 1. In both cases, we consider the heat absorption to be purely superficial; in other words, we are neglecting any sort of penetration of the pump and probe lasers. Under these assumptions, Eq. (5) gives

$$\nabla^2 T(\mathbf{r}, \omega) - \sigma_{cv}^2 T(\mathbf{r}, \omega) = -\frac{1}{k} F(r, \omega)(1 + i\omega\tau)\delta(z), \quad (6)$$

where the pump and probe beams are considered to be Gaussian spots, which induce a superficial heat flux described by

$$F(r, \omega) = \frac{F_0}{2} e^{-r^2/W^2}(1 + e^{i\omega t}), \quad (7)$$

being $F_0(\lambda_{ext})$ the amplitude of the heat absorbed by the sample at excitation wavelength λ_{ext}, W is the full-width half max of the Gaussian spot, and ω is the modulation frequency of the laser intensity. We consider the equation for the Green function problem in cylindrical coordinates with azimuthal symmetry associated with Eq. (6),

$$\frac{1}{r}\frac{\partial}{\partial r}\left[r\frac{\partial}{r}G(r, z; r', z'; \omega)\right] + \frac{\partial^2}{\partial z^2}G(r, z; r', z'; \omega) - \sigma_{cv}^2 G(r, z; r', z'; \omega) = -\frac{1}{2\pi r k}\delta(r - r')\delta(z - z'). \quad (8)$$

Due to the symmetry of the system, we can reduce the 3D problem into a 1D problem by taking the Hankel transform of Eq. (8), defined as

$$g(\lambda, z; r_0, z_0; \omega) = \int_0^\infty G(r, z; r_0, z_0; \omega) J_0(\lambda r)\, r\, dr. \quad (9)$$

In this equation, λ is the variable in Hankel space ($r \to \lambda$).

A. Half-space

We first consider the case of an infinite half-space consisting of a SiC. In this section, we take $\tau_{SiC} = 10^{-8}$ s, which is of the order but not equal to reported phonon relaxation times.[50] The thermal properties of all the materials used for this work are shown in Table 1 of the supplementary material. The heat flux is caused by the absorption of a Gaussian laser source at the interface $z = 0$, as depicted in Fig. 1. For light absorption only at the interface, the Hankel transform (see the supplementary material) of the temperature distribution can be found using

$$\begin{aligned} T_H^{\text{halfspace}}(\lambda, z; \omega) &= \frac{F_0}{2k}\frac{e^{-\varepsilon}}{\varepsilon}\int_0^\infty e^{(-r_0/W)^2} g(\lambda, z|r_0, z = 0; \omega) r_0 dr_0 \\ &= \frac{F_0 W^2}{k}\frac{e^{-\varepsilon(\lambda,\omega)z}}{\varepsilon(\lambda, \omega)} e^{-\lambda^2 W^2/4}(1 + i\omega\tau), \end{aligned} \quad (10)$$

where $\varepsilon(\lambda, \omega) = \sqrt{\lambda^2 + \sigma^2(\omega)}$. The inverse Hankel transform gives the temperature distribution in real space,

$$T(r; \omega) = \left(\frac{F_0 W^2}{k}\right)(1 + i\omega\tau)\int_0^\infty \frac{1}{\varepsilon(\lambda, \omega)} e^{-\varepsilon(\lambda,\omega)z - \lambda^2 W^2/4} J_0(\lambda r)\lambda d\lambda. \quad (11)$$

In Eq. (11), the temperature profile is complex and can be written as $T(r, z = 0, \omega) = |T|e^{i\phi}$. Notice that the CV and Fourier equations are solved using the same method, where one or the other can be found simply by using the corresponding expression of σ and the Fourier transport is recovered in the limit $\tau \to 0$.

The predicted temperature distribution at the interface $z = 0$ is presented in Fig. 2 for the CV and Fourier models. Panels (a) and (b) show the color map for the temperature distribution as a function of frequency, and the distance from the center of the Gaussian laser spot is shown to be normalized to the full-width at half-maximum of the laser spot r/W_0. Panel (c) shows $|\Delta T|$ as a function of frequency at the spatial cut indicated by the dashed lines in (a). The effect of Cattaneo's relaxation time becomes relevant as ω approach to τ^{-1}. Panel (d) shows $|\Delta T|$ as a function of r/W_0 at the frequency cut by the solid line. In this case, the behavior of Fourier and Cattaneo is different only very close to the point

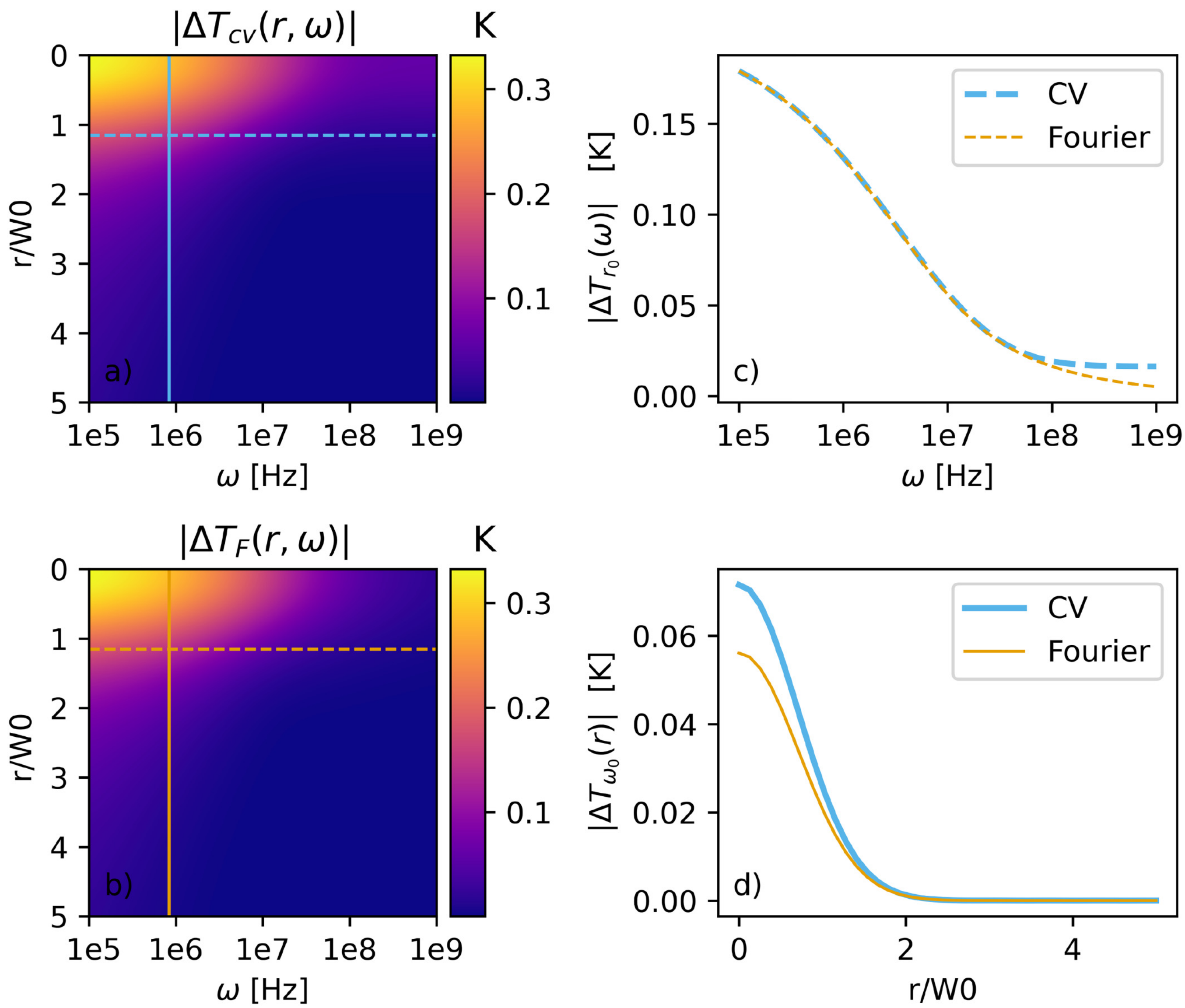

FIG. 2. (a) and (b) show the temperature for SiC half-space with $\tau_{SiC} = 10^{-8}$ s, vertical and horizontal lines on the temperature and phase maps show the "cuts" of the (c) spectral and (d) spatial distribution plots where the predicted temperature is compared between the CV and the Fourier models.

of maximum laser intensity; outside that region, they are undistinguishable. The location of the dashed lines in (a) and (c) was chosen at a value of r/w_0 and frequency where the difference between the Fourier and CV models is more evident. At larger values of r/w_0, the temperature difference between both models becomes negligible.

In Fig. 3, we show the predicted weighted phase of the SiC half-space system for Fourier's and Cattaneo's models. The averaging of the phase models the measured phase lag in the TR experiment.[51] Panel (a) shows the phase behavior, and it is similar for low frequencies in both models where the source induces the phase appearance; again, the effect of Cattaneo's relaxation time becomes relevant as ω approach to τ^{-1}. When the frequency increases, we observe that Fourier's solution loses the frequency dependence, and the phase approaches $-\pi/4$, in contrast to Cattaneo's phase, where we observe a strong dependence on frequency due to the influence of its response time. FDTR data are typically fitted to a model to extract thermal parameters; with this in mind, in panel (b), we show the sensitivity of the phase on the time lag parameter τ, where the sensitivity of the

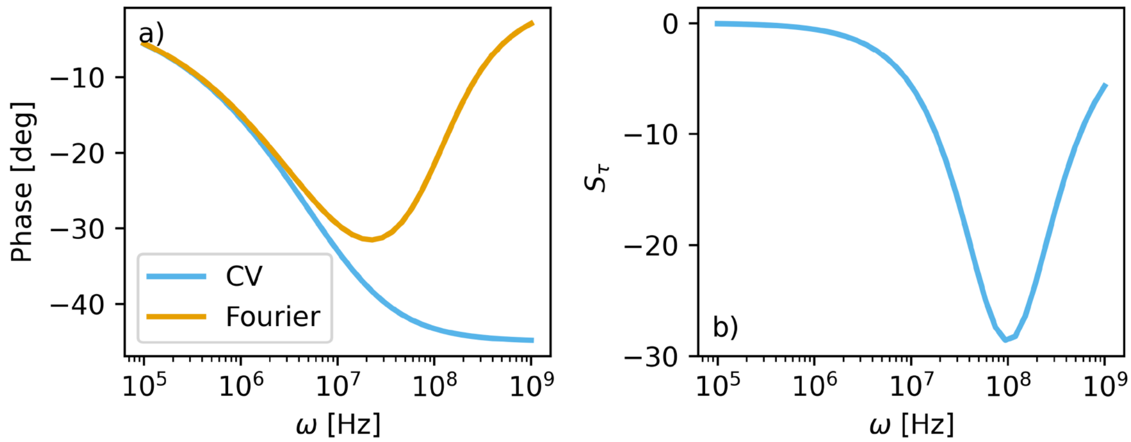

FIG. 3. (a) Weighted phase and (b) τ sensitivity for the SiC half-space system, with $\tau = 10^{-8}$ s.

measurement is given by $S_\tau = \frac{d\phi}{dln\tau}$.[51] We observe that FDTR shows significant sensitivity for frequencies at the τ^{-1} threshold; therefore, it will be possible to fit for the time-relaxation parameter in this frequency range.

B. Layered system

A more realistic model for a typical TR experiment consists of a layered system. Frequently, a thin transducer layer of a highly absorbing metal, such as gold, may be added over the sample. The interaction of the thermal waves due to the interfaces is expected to further enhance non-Fourier-like phenomena due to transmission, reflection, and/or interference of thermal waves. As in the case of the half-space, we consider the absorption of the pump laser to occur entirely at the surface of the transducer layer.

The solution for the temperature field at the surface of the transducer in Hankel space is obtained using the transfer matrix method,[51] given by

$$T_H^{\text{layered}}(\lambda, z; \omega) = (1 + i\omega\tau)\frac{F_0}{k\varepsilon_1(\omega)} W^2 e^{-\lambda^2 W^2/4} A_0 \left(e^{\varepsilon_1^{cv}(\omega)z} + e^{-\varepsilon_1^{cv}(\omega)z}\right), \tag{12}$$

where $\varepsilon_i(\lambda, \omega) = \sqrt{\lambda^2 + \sigma_i^2(\omega)}$, and the subscript i indicates the layer number. The coefficient A_0 is given by

$$A_0(\lambda, z, \omega) = \frac{e^{-\varepsilon_1 z_0}}{2k_1\varepsilon_1}\left(\frac{e^{\varepsilon_1 z_0}[\mathscr{T}]_{12} + e^{-\varepsilon_1 z_0}[\mathscr{T}]_{22}}{e^{\varepsilon_1 z_0}[\mathscr{T}]_{12} - e^{-\varepsilon_1 z_0}[\mathscr{T}]_{22}}\right), \tag{13}$$

where $[\mathscr{T}]_{ij}$ is the (i, j) entry of the associated transfer matrix, resulting in the recursive application of the boundary conditions at each layer. In our system, the factor $A_0(z_0)$ is evaluated at $z_0 = 0$. The form of A_0 is similar to the reciprocal of a reflection coefficient. A more detailed derivation, as well as the expression for the transfer matrix, is found in the supplementary material.

The temperature in real space can be found from Eq. (12) by carrying out the inverse Hankel transform, $T(r, z; \omega) = \int_0^\infty T_H(\lambda, z; \omega) J_0(\lambda r)\lambda d\lambda$. The temperature profile on the top interface of the transducer layer $z = z_0 = 0$ is

$$T(r, 0; \omega) = 2(1 + i\omega\tau)F_0\, k^{-1} W^2 \int_0^\infty A_0(\lambda, \omega)\varepsilon_1^{-1}(\lambda, \omega)\exp \times (-\lambda^2 W^2/4) J_0(\lambda r)\,\lambda\, d\lambda. \tag{14}$$

Notice that the solutions for the layered and half-space system are very similar except for the effective transmission coefficient A_0.

As a case study, we consider a $1\,\mu$m SiC layer over a Si substrate with a 100 nm Au transducer, a $10\,\mu$m full-width half-maximum Gaussian spot, and an energy transfer of the laser to the system of 1 mW. We take $\tau_{SiC} = 10^{-8}$ s, which is of the order but not equal to the phonon relaxation time.[50] With this in mind, we explore the effect of relaxation times τ_{SiC} within the range of reported phonon relaxation times. We take the Au transducer and the Si substrate to act as Fourier like materials so that only SiC has a non-negligible response time. In Fig. 4, we show the predicted temperature and the phase spectrum $T(r, z, \omega) = |T|e^{i\phi}$ of the Au/SiC/Si system on the transducer surface $z = 0$ using the CV and Fourier models. Similar to the half-space system, the absolute value of the temperature is very similar for both models as shown in panels (a) and (b). However, at frequencies near τ^{-1}, there is a strong phase lag region not seen in the Fourier model, panels (c) and (d). The overall behavior is summarized in panels (e) and (f), which show the sensitivity for the CV model and the phase lag between the CV and Fourier models.

To understand the behavior of the temperature profile, we identify three regimes in the frequency space; in the low-frequency regime ($\omega << \tau^{-1}$), the phase ϕ remains approximately constant with no significant differences between models suggesting that in this range, the dynamic is primarily governed by Fourier's law as

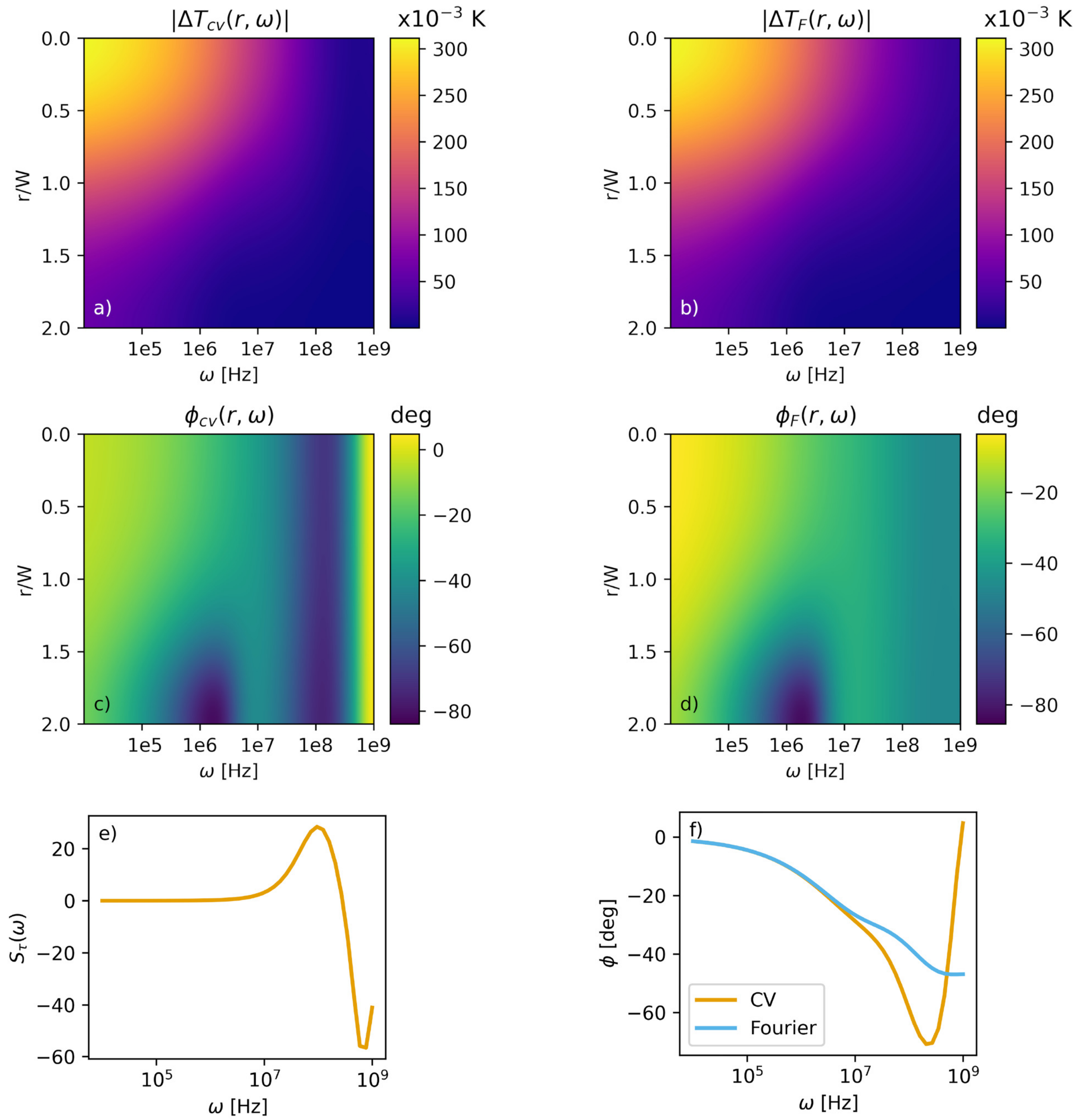

FIG. 4. Predicted temperature [(a) and (b)] and phase [(c) and (d)] maps for the Au/SiC/Si system. The left column corresponds to the CV case and the right one to Fourier behavior. Panels (e) and (f) show the sensitivity of the measurement with respect to the relaxation-time parameter and the weighted phase.

can be seen in Figs. 3 and 4. The transition regime ($\omega \sim \tau^{-1}$), where as the frequency approaches τ^{-1}, the relaxation time τ becomes more relevant leading to changes in the phase. For the case of the layered system, the real and imaginary components of A_0, which involve the thermal wave number σ, define the behavior.

Finally, in the high-frequency regime ($\omega >> \tau^{-1}$), we can use Eqs. (11) and (14) to obtain the phase limit,

$$\phi \approx -\frac{\pi}{4} + \frac{1}{2}\arctan(\omega\tau) + \arctan(\mathrm{Im}A_0/\mathrm{Re}A_0). \tag{15}$$

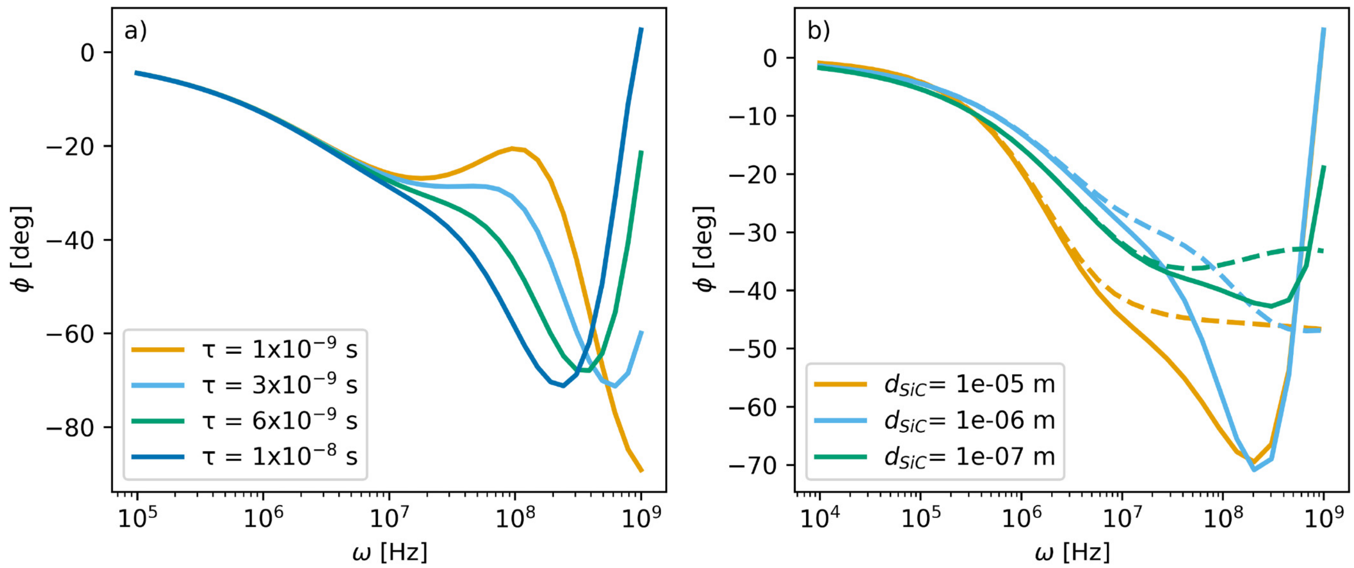

FIG. 5. Predicted phase for the Au/SiC/Si system. (a) The value of τ is varied for a fixed layer width $d_{SiC} = 10^{-6}$ m. (b) Varying the SiC layer width for $\tau = 10^{-8}$ s. Dotted lines are for the Fourier model, while solid lines correspond to the CV model.

From this equation, we see that changes in the phase will be more noticeable at high frequencies as was noticed since Figs. 3 and 4. Notice that the third term of the previous equation does not exist for the half-space, so the value of the relaxation time could be determined through the phase alone. However, in the high-frequency limit, the arctan $(\omega\tau)$ asymptotically approaches $\pi/2$, and this leads to cancellation of the Fourier phase.

To further elucidate the role of the relaxation time, in Fig. 5(a), we show the weighted average of the phase over the Gaussian spot of the probe beam for a fixed thickness of the SiC. Here, we assume that the spot probe and the pump beam are of equal dimension. In this frequency range, relaxation times smaller than 10^{-9} s are practically identical and represent the Fourier limit of the system. First, we point out that the phase predicted by the CV model tends to that of the Fourier model as τ tends to zero concerning the exciting frequency. However, deviations from the Fourier model are apparent even for relaxation times of 3×10^{-9}. We point out that for this value of τ, the differences between the Fourier and non-Fourier description are outside of the frequency range of common FDTR setups. While the analytical analysis is valid, FDTR capabilities will need to increase to reach this frequency range. The FDTR technique can describe the dissimilarities between diffusive and non-diffusive heat transport. In Fig. 5(b), a non-trivial dependence on the width of the SiC layer can be seen by keeping the value of τ fixed. Deviations between CV and Fourier models become apparent at larger frequencies for the smallest layer width. However, there is no apparent direct relation between the layer width and deviations from the models, as $d_{siC} = 10^{-6}$ m shows the largest deviations and would be the ideal case to find differences between CV and Fourier models. The dependence of the phase ϕ with the layer width enters through the transfer matrix when applying the boundary conditions and is not straightforward; however, it can be seen that optimal values of the system width can be found to highlight the difference between the CV and Fourier response.

C. Temperature crystal

Similar to the case of photonic crystals, past works have predicted bandgaps in layered systems consisting of semiconductors and biological-like materials. We, therefore, consider the TR measurements on a Bragg thermal mirror.[11] Although the frequency range of typical FDTR systems is in the order of tens of hertz and is outside the range of typical working conditions of FDTR, we use the same parameters as in the past work for the sake of comparison. In such a system, discrepancies between Fourier and CV models should be most noticeable. We consider a Bragg thermal mirror consisting of a 100 nm Au transducer and a unit cell consisting of one $1\,\mu$m dermis layer and $1\,\mu$m epidermis with time constants 1 and 2 s, respectively, with each unit cell repeated three times with a Si substrate.

The excitation spectrum is now taken between 0 and 40 Hz, shown in Fig. 6. The differences between CV and Fourier models are now noticeable in the temperature distribution [panels (a) and (b)] and in the phase [panels (c) and (d)]. Compared to the Fourier model, there is less heating of the transducer surface for the CV model at low frequencies, and a gap is formed at around 30 Hz. As shown in past works,[11] gaps are formed due to the interference effects of the thermal waves between each layer. We point out that at 30 Hz, there is a dip in the temperature distribution even in the spatial region of $r/W < 1$. This is in accordance with the fact that in this spectral window, the temperonic crystal shows high thermal

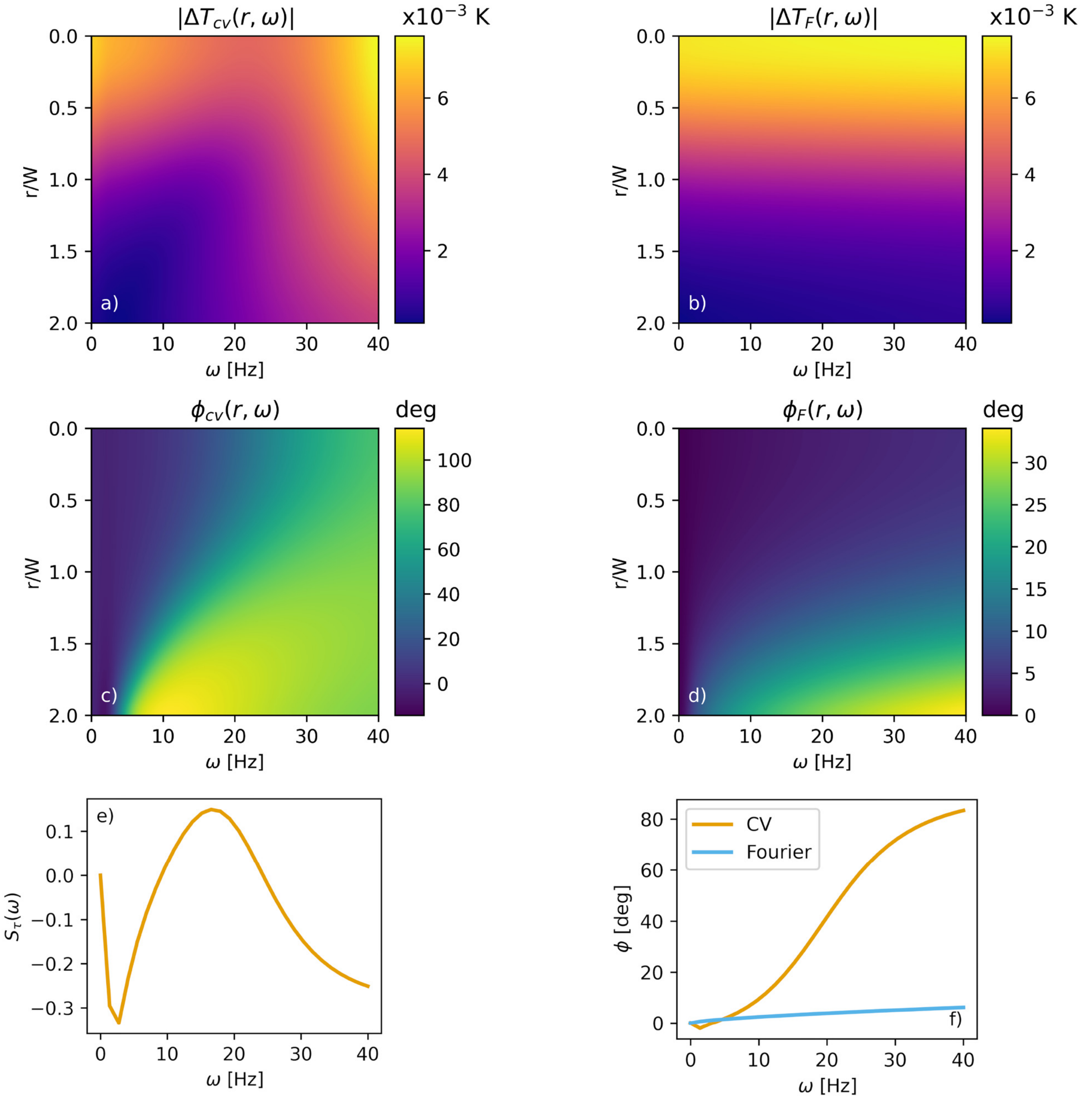

FIG. 6. Thermal Bragg mirror consisting of a 100 nm Au transducer and a unit cell repeated three times. Each unit cell is made of a 1 μm thick dermis layer and a 1 μm thick epidermis layer. Panels (a) and (b) correspond to the temperature distribution for the CV and Fourier models, respectively, and (c) and (d) are the changes in the phase. Panels (e) and (f) show the sensitivity of the measurement with respect to the relaxation-time parameter and the weighted phase.

transitivity. On the other hand, at 40 Hz, the temperature in regions $r/W > 1$ is comparable to that where $r/W < 1$ so that the temperature distribution is not so concentrated near the laser spot, hinting that thermal energy is not so easily transmitted in the z direction at this frequency.

IV. CONCLUSION

This study demonstrates a robust methodology for determining relaxation times in non-Fourier materials using thermoreflectance techniques. The high sensitivity and resolution in the phase measurements of just a few millidegrees make the technique ideal to distinguish between Fourier and non-Fourier behavior of the materials. By leveraging the pump-probe thermoreflectance approach, we effectively captured high-resolution transient thermal responses, enabling precise fitting to a non-Fourier heat conduction model based on the Cattaneo–Vernotte equation. Our findings reveal that thermoreflectance is a powerful tool for characterizing the delayed heat flux response in materials, allowing for accurate quantification of relaxation times that govern non-Fourier behavior. The results underscore the sensitivity of thermoreflectance techniques to nanoscale thermal transport parameters, particularly in materials where traditional Fourier heat conduction models fail to capture rapid temperature changes. By pointing out the differences in the predicted phase lags between Fourier and CV models, this work can help improve the interpretation of FDTR results and pave the wave for systematic characterization of time-relaxation parameters in non-Fourier material. This work highlights the importance of considering relaxation-time effects in the design of thermal management systems for microelectronics, optoelectronics, and other advanced technologies, where efficient heat dissipation at the nanoscale is critical.

The system described in this work makes the assumption that roughness is negligible as compared to the wavelength of the probe laser. In an actual experiment, surface roughness has to be measured since it can be a confounder during measurements when working with reflectance measurements.[52,53] Roughness can induce a phase change in the measurement not related to a non-Fourier behavior, which can be worth of study in a future work. The approach presented in this paper can be extended to a broader range of materials and thermal excitation conditions, further validating the versatility and applicability of thermoreflectance-based relaxation-time determination. Additionally, exploring the impact of varying relaxation times in composite and anisotropic materials could deepen our understanding of non-Fourier heat transfer mechanisms across different material classes. Overall, this study provides a foundational technique for analyzing and optimizing thermal behavior in emerging materials where non-Fourier effects play a significant role.

SUPPLEMENTARY MATERIAL

A table of the thermal properties of the materials used in this work, the associated Green function equation in cylindrical coordinates, as well as a derivation of the equations of the layered system, in particular, the transmission coefficient equation (13), can be found in the supplementary material.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

A. Camacho de la Rosa: Conceptualization (equal); Formal analysis (equal); Methodology (equal); Writing – original draft (equal). **R. Esquivel-Sirvent:** Conceptualization (equal); Formal analysis (equal); Methodology (equal); Writing – original draft (equal). **D. Becerril:** Conceptualization (equal); Formal analysis (equal); Methodology (equal); Writing – original draft (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

REFERENCES

[1] T. N. Narasimhan, "Fourier's heat conduction equation: History, influence, and connections," Rev. Geophys. **37**, 151, https://doi.org/10.1029/1998RG900006 (1999).
[2] O. M. Necati *et al.*, *Heat Conduction* (Wiley, 1993).
[3] J. Luciani, P. Mora, and J. Virmont, "Nonlocal heat transport due to steep temperature gradients," Phys. Rev. Lett. **51**, 1664 (1983).
[4] M. J. Maurer, "Relaxation model for heat conduction in metals," J. Appl. Phys. **40**, 5123 (1969).
[5] R. Anufriev, A. Ramiere, J. Maire, and M. Nomura, "Heat guiding and focusing using ballistic phonon transport in phononic nanostructures," Nat. Commun. **8**, 15505 (2017).
[6] C. Cattaneo, *Some Aspects of Diffusion Theory*, edited by A. Pignedoli (Springer, Berlin, 2011), pp. 485–485.
[7] P. Vernotte, "Les paradoxes de la théorie continue de lequation de la chaleur," C. R. Acad. Sci. **246**, 3154 (1958).
[8] D. D. Joseph and L. Preziosi, "Heat waves," Rev. Mod. Phys. **61**, 41 (1989).
[9] A. Kabiri and M. R. Talaee, "Analysis of hyperbolic Pennes bioheat equation in perfused homogeneous biological tissue subject to the instantaneous moving heat source," SN Appl. Sci. **3**, 398 (2021).
[10] M. Farhat, S. Guenneau, P. Y. Chen, A. Alù, and K. N. Salama, "Scattering cancellation-based cloaking for the Maxwell-Cattaneo heat waves," Phys. Rev. Appl. **11**, 044089 (2019).
[11] A. Camacho de la Rosa, D. Becerril, M. G. Gómez-Farfán, and R. Esquivel-Sirvent, "Bragg mirrors for thermal waves," Energies **14**, 7452 (2021).
[12] M. Gandolfi, G. Benetti, C. Glorieux, C. Giannetti, and F. Banfi, "Accessing temperature waves: A dispersion relation perspective," Int. J. Heat Mass Transf. **143**, 118553 (2019).
[13] A. C. de la Rosa and R. Esquivel-Sirvent, "Causality in non-Fourier heat conduction," J. Phys. Commun. **6**, 105003 (2022).
[14] S. Sobolev, "Discrete heat conduction equation: Dispersion analysis and continuous limits," Int. J. Heat Mass Transf. **221**, 125062 (2024).
[15] D. Y. Tzou, "Lagging behavior in other transport processes," in *Macro- to Microscale Heat Transfer* (John Wiley & Sons, Ltd., 2014), Chap. 9, pp. 327–357.
[16] J. Ordóñez-Miranda and J. Alvarado-Gil, "Exact solution of the dual-phase-lag heat conduction model for a one-dimensional system excited with a periodic heat source," Mech. Res. Commun. **37**, 276 (2010).
[17] D. Y. Tzou, *Macro-to Microscale Heat Transfer: The Lagging Behavior* (John Wiley & Sons, 2014).
[18] D. Y. Tzou and J. K. Chen, "Thermal lagging in random media," J. Thermphys. Heat Transf. **12**, 567 (1998).

[19] R. Kovács, "Heat equations beyond Fourier: From heat waves to thermal metamaterials," Phys. Rep. **1048**, 1 (2024).
[20] A. Camacho De La Rosa, D. Becerril, G. Gomez-Farfan, and R. Esquivel-Sirvent, "Time-harmonic photothermal heating by nanoparticles in a non-Fourier medium," J. Phys. Chem. C **125**, 22856 (2021).
[21] A. Banerjee, A. A. Ogale, C. Das, K. Mitra, and C. Subramanian, "Temperature distribution in different materials due to short pulse laser irradiation," Heat Transf. Eng. **26**, 41 (2005).
[22] M. I. Tribelsky, A. E. Miroshnichenko, Y. S. Kivshar, B. S. Luk'yanchuk, and A. R. Khokhlov, "Laser pulse heating of spherical metal particles," Phys. Rev. X **1**, 021024 (2011).
[23] F. Tang, Y. Zhang, J. Zhang, J. Guo, and R. Liu, "Assessment of the efficacy of laser hyperthermia and nanoparticle-enhanced therapies by heat shock protein analysis," AIP Adv. **4**, 031334 (2014).
[24] C. Liu, C. C. Mi, and B. Q. Li, "Transient temperature response of pulsed-laser-induced heating for nanoshell-based hyperthermia treatment," IEEE Trans. Nanotech. **8**, 697 (2009).
[25] A. N. Yakunin, Y. A. Avetisyan, and V. V. Tuchin, "Quantification of laser local hyperthermia induced by gold plasmonic nanoparticles," J. Biomed. Opt. **20**, 1 (2015).
[26] Z. Qin and J. C. Bischof, "Thermophysical and biological responses of gold nanoparticle laser heating," Chem. Soc. Rev. **41**, 1191 (2012).
[27] T.-L. Cao, T.-A. Le, Y. Hadadian, and J. Yoon, "Theoretical analysis for using pulsed heating power in magnetic hyperthermia therapy of breast cancer," Int. J. Mol. Sci. **22**, 8895 (2021).
[28] Z. Ge, Y. Kang, T. A. Taton, P. V. Braun, and D. G. Cahill, "Thermal transport in Au-core polymer-shell nanoparticles," Nano Lett. **5**, 531 (2005).
[29] G. Baffou and R. Quidant, "Thermo-plasmonics: Using metallic nanostructures as nano-sources of heat," Laser Photonics Rev. **7**, 171 (2013).
[30] G. Benenti, D. Donadio, S. Lepri, and R. Livi, "Non-Fourier heat transport in nanosystems," Riv. Nuov. Cim. **46**, 105 (2023).
[31] M. Gandolfi, C. Giannetti, and F. Banfi, "Temperonic crystal: A superlattice for temperature waves in graphene," Phys. Rev. Lett. **125**, 265901 (2020).
[32] C. A. Romero-Ramos, J. Manzanares-Martínez, B. Manzanares-Martínez, D. Soto-Puebla, G. A. Morales-Morales, and C. E. Ruiz-Rosales, "Brewster angle of thermal diffusivity waves at an interface," Results Phys. **30**, 104856 (2021).
[33] G. Morales-Morales and J. Manzanares-Martinez, "Enlargement of band gaps on thermal wave crystals by using heterostructures," Results Phys. **42**, 106019 (2022).
[34] F. Xu, K. Seffen, and T. Lu, "Non-Fourier analysis of skin biothermomechanics," Int. J. Heat Mass Transf. **51**, 2237 (2008).
[35] M. Jaunich, S. Raje, K. Kim, K. Mitra, and Z. Guo, "Bio-heat transfer analysis during short pulse laser irradiation of tissues," Int. J. Heat Mass Transf. **51**, 5511 (2008).
[36] A. V. K. Mitra, S. Kumar, and M. Moallemi, "Experimental evidence of hyperbolic heat conduction in processed meat," J. Heat Transf. **117**, 568 (1995).
[37] A. Madhukar, Y. Park, W. Kim, H. J. Sunaryanto, R. Berlin, L. P. Chamorro, J. Bentsman, and M. Ostoja-Starzewski, "Heat conduction in porcine muscle and blood: Experiments and time-fractional telegraph equation model," J. R. Soc. Interface **16**, 20190726 (2019).
[38] W. Roetzel, N. Putra, and S. K. Das, "Experiment and analysis for non-Fourier conduction in materials with non-homogeneous inner structure," Int. J. Therm. Sci. **42**, 541 (2003).
[39] K. T. Regner, S. Majumdar, and J. A. Malen, "Instrumentation of broadband frequency domain thermoreflectance for measuring thermal conductivity accumulation functions," Rev. Sci. Instrum. **84**, 064901 (2013).
[40] D. J. Kirsch, J. Martin, R. Warzoha, M. McLean, D. Windover, and I. Takeuchi, "An instrumentation guide to measuring thermal conductivity using frequency domain thermoreflectance (FDTR)," Rev. Sci. Instrum. **95**, 103006 (2024).
[41] J. Zhu, D. Tang, W. Wang, J. Liu, K. W. Holub, and R. Yang, "Ultrafast thermoreflectance techniques for measuring thermal conductivity and interface thermal conductance of thin films," J. Appl. Phys. **108**, 094315 (2010).
[42] M. G. Ghossoub, J.-H. Lee, O. T. Baris, D. G. Cahill, and S. Sinha, "Thermal conductivity of confined ultrathin polymers," MRS Online Proc. Lib. **1172**, 1172 (2009).
[43] M. N. Touzelbaev, P. Zhou, R. Venkatasubramanian, and K. Goodson, "Thermal characterization of Bi_2Te_3/Sb_2Te_3 superlattices," J. Appl. Phys. **90**, 763 (2001).
[44] E. Matatagui, A. Thompson, and M. Cardona, "Thermoreflectance in semiconductors," Phys. Rev. **176**, 950 (1968).
[45] J.-L. Auriault, "Cattaneo–Vernotte equation versus Fourier thermoelastic hyperbolic heat equation," Int. J. Eng. Sci. **101**, 45 (2016).
[46] T. M. Tritt, *Thermal Conductivity: Theory, Properties, and Applications* (Springer Science & Business Media, 2005).
[47] C. J. Glassbrenner and G. A. Slack, "Thermal conductivity of silicon and germanium from 3 k to the melting point," Phys. Rev. **134**, A1058 (1964).
[48] D. Li, Y. Wu, P. Kim, L. Shi, P. Yang, and A. Majumdar, "Thermal conductivity of individual silicon nanowires," Appl. Phys. Lett. **83**, 2934 (2003).
[49] G. Romano, K. Esfarjani, D. A. Strubbe, D. Broido, and A. M. Kolpak, "Temperature-dependent thermal conductivity in silicon nanostructured materials studied by the Boltzmann transport equation," Phys. Rev. B **93**, 035408 (2016).
[50] Y. Ezzahri and K. Joulain, "Dynamical thermal conductivity of bulk semiconductor crystals," J. Appl. Phys. **112**, 083515 (2012).
[51] D. G. Cahill, "Analysis of heat flow in layered structures for time-domain thermoreflectance," Rev. Sci. Instrum. **75**, 5119 (2004).
[52] S. Maity, A. Banerjee, and C. Mitra, "Temperature dependence of the reflectance of metals at visible wavelengths," Proc. SPIE **7792**, 77920D (2010).
[53] Y. Wang, J. Y. Park, Y. K. Koh, and D. G. Cahill, "Thermoreflectance of metal transducers for time-domain thermoreflectance," J. Appl. Phys. **108**, 043507 (2010).